MASTER IN ASTROPHYSICS
AND SPACE SCIENCE

TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

Introduction to Numerical Methods for Astrophysics

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Overview

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$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + u \cdot \nabla Q$$

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  - $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

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Laplace Equation

Poisson Equation

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Lid Driven Cavity Flow

Pressure Driven Channel Flow

Nonuniform Periodic Flow

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Thank You!

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Resources on GitHub: here